Year 11 Foundation

# REVISION PACK

## Progress Check 1

**Thursday 1 October 2026**

**Name:** _____

**Class:** _____

### Units in this pack

- Number 8
- Algebra 7

## Number 8 · Standard form (Lesson 1.1)

**Worked example: writing numbers in standard form**

This question is about standard form.

(a) Write $3.8 \times 10^4$ as an ordinary number. (b) Write 0.00065 in standard form.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Multiply by 10 000 | $3.8 \times 10\,000 = 38\,000$ |
| **(b)** | **Step 1** | Point after the 6 | $6.5$ |
| | **Step 2** | Small number: negative power | $6.5 \times 10^{-4}$ |

### Practice: writing numbers in standard form

**1.1** This question is about standard form.

(a) Write $2.7 \times 10^4$ as an ordinary number.

(b) Write 0.0063 in standard form.

**1.2** This question is about standard form.

(a) Write $5.04 \times 10^5$ as an ordinary number.

(b) Write 0.039 in standard form.

**1.3** This question is about standard form.

(a) Write $9.1 \times 10^3$ as an ordinary number.

(b) Write 0.00805 in standard form.

**1.4** This question is about standard form.

(a) Write $6.25 \times 10^3$ as an ordinary number.

(b) Write 0.00051 in standard form.

**1.5** This question is about standard form.

(a) Write $1.8 \times 10^5$ as an ordinary number.

(b) Write 0.0027 in standard form.

**1.6** This question is about standard form.

(a) Write $7.3 \times 10^6$ as an ordinary number.

(b) Write 0.068 in standard form.

## Number 8 · Standard form (Lesson 1.1)

**Worked example: multiplying in standard form**

Work out $(2 \times 10^5) \times (4 \times 10^3)$. Give your answer in standard form.

| | | |
|---|---|---|
| **Step 1** | Multiply the numbers | $2 \times 4 = 8$ |
| **Step 2** | Add the powers | $10^{5+3} = 10^8$ |
| **Step 3** | Answer | $8 \times 10^8$ |

### Practice: multiplying in standard form

**1.7** Work out $(2 \times 10^3) \times (3 \times 10^5)$. Give your answer in standard form.

**1.8** Work out $(3 \times 10^6) \times (3 \times 10^2)$. Give your answer in standard form.

**1.9** Work out $(1.5 \times 10^4) \times (2 \times 10^3)$. Give your answer in standard form.

**1.10** Work out $(4 \times 10^2) \times (2 \times 10^7)$. Give your answer in standard form.

**1.11** Work out $(2.5 \times 10^3) \times (3 \times 10^4)$. Give your answer in standard form.

**1.12** Work out $(1.2 \times 10^5) \times (4 \times 10^6)$. Give your answer in standard form.

## Number 8 · Standard form (Lesson 1.1)

**Worked example: big and small numbers**

This question is about standard form.

(a) Write 7 300 000 in standard form. (b) Write $4.5 \times 10^{-3}$ as an ordinary number.

| | | | |
|---|---|---|---|
| (a) | Step 1 | Point after the 7 | $7.3$ |
| | Step 2 | Count 6 places | $7.3 \times 10^{6}$ |
| (b) | Step 1 | Divide by 1000 | $4.5 \div 1000 = 0.0045$ |

### Practice: big and small numbers

**1.13** This question is about standard form.

(a) Write 3 800 000 in standard form.

(b) Write $2.6 \times 10^{-2}$ as an ordinary number.

**1.14** This question is about standard form.

(a) Write 65 000 in standard form.

(b) Write $7.1 \times 10^{-3}$ as an ordinary number.

**1.15** This question is about standard form.

(a) Write 920 000 000 in standard form.

(b) Write $5.08 \times 10^{-4}$ as an ordinary number.

**1.16** This question is about standard form.

(a) Write 410 000 in standard form.

(b) Write $1.9 \times 10^{-2}$ as an ordinary number.

**1.17** This question is about standard form.

(a) Write 28 000 000 in standard form.

(b) Write $6.45 \times 10^{-3}$ as an ordinary number.

**1.18** This question is about standard form.

(a) Write 5300 in standard form.

(b) Write $8 \times 10^{-5}$ as an ordinary number.

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: prime factors**

Write 84 as a product of its prime factors.

| | | |
|---|---|---|
| **Step 1** | Divide by 2 | $84 = 2 \times 42$ |
| **Step 2** | Divide by 2 | $42 = 2 \times 21$ |
| **Step 3** | Divide by 3 | $21 = 3 \times 7$ |
| **Step 4** | Answer | $2 \times 2 \times 3 \times 7$ |

### Practice: prime factors

**2.1** Write 90 as a product of its prime factors.

**2.2** Write 126 as a product of its prime factors.

**2.3** Write 150 as a product of its prime factors.

**2.4** Write 48 as a product of its prime factors.

**2.5** Write 200 as a product of its prime factors.

**2.6** Write 132 as a product of its prime factors.

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: HCF and LCM**

Find

(a) the highest common factor (HCF) of 18 and 30 (b) the lowest common multiple (LCM) of 4 and 10

| | | | |
|---|---|---|---|
| (a) | Step 1 | Factors of 18 | $1, 2, 3, 6, 9, 18$ |
| | Step 2 | Factors of 30 | $1, 2, 3, 5, 6, 10, 15, 30$ |
| | Step 3 | Answer | $6$ |
| (b) | Step 1 | Multiples of 4 | $4, 8, 12, 16, 20$ |
| | Step 2 | Multiples of 10 | $10, 20$ |
| | Step 3 | Answer | $20$ |

### Practice: HCF and LCM

**2.7** Find

(a) the highest common factor (HCF) of 20 and 30 (b) the lowest common multiple (LCM) of 4 and 6

**2.8** Find

(a) the highest common factor (HCF) of 16 and 40 (b) the lowest common multiple (LCM) of 6 and 9

**2.9** Find

(a) the highest common factor (HCF) of 27 and 45 (b) the lowest common multiple (LCM) of 8 and 12

**2.10** Find

(a) the highest common factor (HCF) of 28 and 42 (b) the lowest common multiple (LCM) of 5 and 7

**2.11** Find

(a) the highest common factor (HCF) of 32 and 48 (b) the lowest common multiple (LCM) of 9 and 12

**2.12** Find

(a) the highest common factor (HCF) of 15 and 35 (b) the lowest common multiple (LCM) of 10 and 15

## Number 8 · Factors, multiples and primes (Lessons 1.2, 1.3)

**Worked example: prime factors and the HCF**

This question is about 90 and 48

(a) Write 90 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 90 and 48

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Divide by primes | $90 = 2 \times 45 = 2 \times 3 \times 15$ |
| | **Step 2** | Answer | $2 \times 3 \times 3 \times 5$ |
| **(b)** | **Step 1** | Prime factors of 48 | $2 \times 2 \times 2 \times 2 \times 3$ |
| | **Step 2** | Primes in both | $2 \times 3 = 6$ |

### Practice: prime factors and the HCF

**2.13** This question is about 72 and 54

(a) Write 72 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 72 and 54

**2.14** This question is about 120 and 90

(a) Write 120 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 120 and 90

**2.15** This question is about 126 and 84

(a) Write 126 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 126 and 84

**2.16** This question is about 100 and 75

(a) Write 100 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 100 and 75

**2.17** This question is about 60 and 45

(a) Write 60 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 60 and 45

**2.18** This question is about 98 and 70

(a) Write 98 as a product of its prime factors.

(b) Find the highest common factor (HCF) of 98 and 70

## Number 8 · Ratio (Lesson 1.4)

**Worked example: simplifying ratios**

Write the ratio 20 : 35 in its simplest form.

| | | |
|---|---|---|
| **Step 1** | HCF of 20 and 35 | $5$ |
| **Step 2** | Divide both by 5 | $20 \div 5 = 4,\ 35 \div 5 = 7$ |
| **Step 3** | Answer | $4 : 7$ |

### Practice: simplifying ratios

**3.1** Write the ratio 15 : 25 in its simplest form.

**3.2** Write the ratio 28 : 42 in its simplest form.

**3.3** Write the ratio 36 : 48 in its simplest form.

**3.4** Write the ratio 30 : 45 in its simplest form.

**3.5** Write the ratio 16 : 40 in its simplest form.

**3.6** Write the ratio 27 : 63 in its simplest form.

## Number 8 · Ratio (Lesson 1.4)

**Worked example: sharing in a ratio**

Ali and Sara share 56 KD in the ratio 3 : 4. How much does each person get?

| | | |
|---|---|---|
| **Step 1** | Add the parts | $3 + 4 = 7$ |
| **Step 2** | One part | $56 \div 7 = 8$ |
| **Step 3** | Ali | $3 \times 8 = 24$ KD |
| **Step 4** | Sara | $4 \times 8 = 32$ KD |

### Practice: sharing in a ratio

**3.7** Omar and Mona share 60 KD in the ratio 2 : 3. How much does each person get?

**3.8** Fahad and Laila share 72 KD in the ratio 5 : 3. How much does each person get?

**3.9** Noor and Ahmad share 90 KD in the ratio 7 : 2. How much does each person get?

**3.10** Yousef and Huda share 40 KD in the ratio 3 : 5. How much does each person get?

**3.11** Sara and Ali share 84 KD in the ratio 5 : 7. How much does each person get?

**3.12** Maha and Faisal share 100 KD in the ratio 3 : 7. How much does each person get?

## Number 8 · Ratio (Lesson 1.4)

**Worked example: recipes**

In a recipe, the ratio of flour to sugar is 4 : 3. Mona uses 280 g of flour. How much sugar should Mona use?

| | | |
|---|---|---|
| **Step 1** | One part | $280 \div 4 = 70$ |
| **Step 2** | Sugar is 3 parts | $3 \times 70 = 210$ |
| **Step 3** | Answer | $210\ \text{g}$ |

### Practice: recipes

**3.13** In a recipe, the ratio of flour to butter is 3 : 2. Ali uses 240 g of flour. How much butter should Ali use?

**3.14** In a recipe, the ratio of flour to sugar is 7 : 2. Sara uses 420 g of flour. How much sugar should Sara use?

**3.15** In a recipe, the ratio of juice to water is 2 : 5. Omar uses 150 ml of juice. How much water should Omar use?

**3.16** In a recipe, the ratio of flour to butter is 4 : 1. Huda uses 360 g of flour. How much butter should Huda use?

**3.17** In a recipe, the ratio of sugar to flour is 3 : 5. Fahad uses 120 g of sugar. How much flour should Fahad use?

**3.18** In a recipe, the ratio of flour to sugar is 5 : 4. Laila uses 450 g of flour. How much sugar should Laila use?

## Number 8 · Ratio (Lesson 1.4)

**Worked example: finding the total**

The ratio of red beads to blue beads in a box is 2 : 5. There are 14 red beads. How many red beads and blue beads are in the box altogether?

| | | |
|---|---|---|
| **Step 1** | One part | $14 \div 2 = 7$ |
| **Step 2** | Total parts | $2 + 5 = 7$ |
| **Step 3** | Answer | $7 \times 7 = 49$ |

### Practice: finding the total

**3.19** The ratio of red beads to blue beads in a box is 3 : 5. There are 18 red beads. How many red beads and blue beads are in the box altogether?

**3.20** The ratio of green counters to yellow counters in a box is 4 : 7. There are 28 green counters. How many green counters and yellow counters are in the box altogether?

**3.21** The ratio of white tiles to black tiles in a box is 2 : 9. There are 12 white tiles. How many white tiles and black tiles are in the box altogether?

**3.22** The ratio of pens to pencils in a box is 5 : 3. There are 35 pens. How many pens and pencils are in the box altogether?

**3.23** The ratio of apples to oranges in a box is 3 : 8. There are 27 apples. How many apples and oranges are in the box altogether?

**3.24** The ratio of red cubes to blue cubes in a box is 4 : 5. There are 32 red cubes. How many red cubes and blue cubes are in the box altogether?

## Algebra 7 · Formulae (Lessons 2.1, 2.2)

**Worked example: substitution into P = 2l + 2w**

$P = 2l + 2w$. Work out the value of $P$ when $l = 9$ and $w = 5$

| | | |
|---|---|---|
| **Step 1** | Put the numbers in | $P = 2 \times 9 + 2 \times 5$ |
| **Step 2** | Multiply first | $= 18 + 10$ |
| **Step 3** | Answer | $P = 28$ |

### Practice: substitution into P = 2l + 2w

**4.1** $P = 2l + 2w$. Work out the value of $P$ when $l = 8$ and $w = 5$

**4.2** $P = 2l + 2w$. Work out the value of $P$ when $l = 12$ and $w = 7$

**4.3** $P = 2l + 2w$. Work out the value of $P$ when $l = 3.5$ and $w = 2$

**4.4** $P = 2l + 2w$. Work out the value of $P$ when $l = 10$ and $w = 4.5$

**4.5** $P = 2l + 2w$. Work out the value of $P$ when $l = 15$ and $w = 9$

**4.6** $P = 2l + 2w$. Work out the value of $P$ when $l = 6$ and $w = 2.5$

## Algebra 7 · Formulae (Lessons 2.1, 2.2)

**Worked example: substitution into v = u + at**

$v = u + at$. Work out the value of $v$ when $u = 8$, $a = 2$ and $t = 6$

| | | |
|---|---|---|
| **Step 1** | Put the numbers in | $v = 8 + 2 \times 6$ |
| **Step 2** | Multiply first | $= 8 + 12$ |
| **Step 3** | Answer | $v = 20$ |

### Practice: substitution into v = u + at

**4.7** $v = u + at$. Work out the value of $v$ when $u = 10$, $a = 4$ and $t = 3$

**4.8** $v = u + at$. Work out the value of $v$ when $u = 5$, $a = 6$ and $t = 7$

**4.9** $v = u + at$. Work out the value of $v$ when $u = 15$, $a = 2$ and $t = 9$

**4.10** $v = u + at$. Work out the value of $v$ when $u = 3$, $a = 5$ and $t = 8$

**4.11** $v = u + at$. Work out the value of $v$ when $u = 20$, $a = 3$ and $t = 4$

**4.12** $v = u + at$. Work out the value of $v$ when $u = 7$, $a = 9$ and $t = 2$

## Algebra 7 · Formulae (Lessons 2.1, 2.2)

**Worked example: using a formula**

The cost, $T$ KD, of hiring a car for $n$ hours is $T = 30n + 20$. Mona pays 140 KD. For how many hours did Mona hire the car?

| | | |
|---|---|---|
| **Step 1** | Make an equation | $30n + 20 = 140$ |
| **Step 2** | Subtract 20 | $30n = 120$ |
| **Step 3** | Divide by 30 | $n = 4$ |
| **Step 4** | Answer | $4$ hours |

### Practice: using a formula

**4.13** The cost, $T$ KD, of hiring a car for $n$ hours is $T = 35n + 15$. Ali pays 155 KD. For how many hours did Ali hire the car?

**4.14** The cost, $T$ KD, of hiring a bike for $n$ hours is $T = 25n + 10$. Omar pays 135 KD. For how many hours did Omar hire the bike?

**4.15** The cost, $T$ KD, of hiring a boat for $n$ hours is $T = 50n + 30$. Fahad pays 280 KD. For how many hours did Fahad hire the boat?

**4.16** The cost, $T$ KD, of hiring a van for $n$ hours is $T = 45n + 20$. Sara pays 200 KD. For how many hours did Sara hire the van?

**4.17** The cost, $T$ KD, of hiring a jet ski for $n$ hours is $T = 60n + 40$. Laila pays 340 KD. For how many hours did Laila hire the jet ski?

**4.18** The cost, $T$ KD, of hiring a kayak for $n$ hours is $T = 20n + 15$. Yousef pays 135 KD. For how many hours did Yousef hire the kayak?

**Worked example: rearranging formulae**

Rearrange each formula to make the given letter the subject.

(a) $y = 3x + 5$ Make $x$ the subject.

(b) $A = \pi r l$ Make $l$ the subject.

| | | | |
|---|---|---|---|
| **(a)** | **Step 1** | Subtract 5 | $y - 5 = 3x$ |
| | **Step 2** | Divide by 3 | $x = \dfrac{y-5}{3}$ |
| **(b)** | **Step 1** | Divide by $\pi r$ | $l = \dfrac{A}{\pi r}$ |

## Practice: rearranging formulae

**4.19** Rearrange each formula to make the given letter the subject.

(a) $y = 5x + 2$ Make $x$ the subject.

(b) $A = lw$ Make $w$ the subject.

**4.20** Rearrange each formula to make the given letter the subject.

(a) $y = 2x - 7$ Make $x$ the subject.

(b) $V = Ah$ Make $h$ the subject.

**4.21** Rearrange each formula to make the given letter the subject.

(a) $y = 6x + 1$ Make $x$ the subject.

(b) $C = \pi d$ Make $d$ the subject.

**4.22** Rearrange each formula to make the given letter the subject.

(a) $y = 3x - 4$ Make $x$ the subject.

(b) $P = 4s$ Make $s$ the subject.

**4.23** Rearrange each formula to make the given letter the subject.

(a) $y = 8x + 3$ Make $x$ the subject.

(b) $F = ma$ Make $a$ the subject.

**4.24** Rearrange each formula to make the given letter the subject.

(a) $y = 4x - 9$ Make $x$ the subject.

(b) $D = st$ Make $t$ the subject.